

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Centre of Excellence in Artificial Intelligence

2. **Subject Name:** Machine Learning Laboratory

L-T-P: 0-0-3

Credits: 2

3. **Pre-requisites:** None

4. **Syllabus and reference books:**

Syllabus:

Exploring feature vectors, train-test splitting, K-nearest neighbor and variants, Discriminative-ness of features, Decision Tree, Linear and regularized regression, Logistic Regression and Decision Boundary, Linear Separability, Perceptron, Multi-class classification, Dimensionality Reduction, Random Forest and Bagging, Adaboost, Gradient Boosting, Support Vector Machine, Kernel Methods, Hierarchical and Agglomerative Clustering; Density-based clustering; K-means clustering and its Variants; Spectral clustering; Gaussian Mixture Model for clustering; Semi-supervised learning using Self-training, co-training, Graph-based Label propagation

Reference Books:

- 1) Introduction to Machine Learning with Python, by Andreas C. Müller, Sarah Guido
Released September 2016, Publisher(s): O'Reilly Media, Inc.
- 2) Christopher M. Bishop, Pattern Recognition for Machine Learning, Springer (2006)
- 3) R. Duda, P. Hart, D. Stork, Pattern Classification, Wiley (2007)

5. **Lecture-wise break-up:**

Sl. No.	Topic	No. of Lab-Hours
1. K-nearest neighbor and variants	Generation of synthetic data and running K-nearest Neighbors and its simple variants, trial on real dataset	3 hours
2. Discriminative-ness of features, Decision Tree	Ranking of features by entropy/information gain/Gini Index, implementation of Decision Tree on standard dataset (camera sales)	3 hours

3. Linear and regularized regression	Comparing Linear and Regularized (Ridge/LASSO) regression, role of trade-off parameter	3 hours
4. Logistic Regression and Decision Boundary	Predicting popularity of comments using logistic regression, visualization of decision boundary	3 hours
5. Linear Separability, Perceptron	Implementation of perceptron algo on different kinds of synthetic datasets, Application of one-vs-one, one-vs-all to a given dataset	3 hours
6. Dimensionality Reduction	Principal Component Analysis, Linear Discriminant Analysis for both real and synthetic data	3 hours
7. Ensemble Methods	Prediction of rainfall at different locations using ensemble methods, Regression problems using Adaboost and Gradient Boosting	3 hours
8. Support Vector Machine	Protein structure classification using SVM, Standard and user-defined kernel functions for non-linearly separable datasets	3 hours
9. Agglomerative and Density-based Clustering	Comparing various linkage criteria on synthetic and real datasets, Anomaly detection of sensor readings using DB-Scan	3 hours
10. K-means clustering and variants	K-means clustering for different real datasets, choosing correct value of “K”, robust variants of K-means including K-means++, K-median, Kernelized K-means on synthetic datasets	3 hours
11. Gaussian Mixture Models	Soft clustering using GMM applied to speaker diarization, comparison with K-means	3 hours
12. Semi-supervised Learning	social network node classification based on graph-based label propagation using a few labelled nodes	3 hours
Total Number of hours		36 hours